ARUNIMA ANAND

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EDUCATION

M.S. in Data Science | University of California, San Diego — Sep 2025 – Dec 2026 | San Diego, CA
GPA: 3.97
Relevant Coursework: Causal Inference, Statistical Models, Machine Learning, Scalable Data Systems, Responsible ML, Recommender Systems

Bachelor's in Economics (Minor in Mathematics) | University of Delhi — Jul 2019 – Jul 2022 | Delhi, India
GPA: 3.88 (First Division with Honors)

PROFESSIONAL EXPERIENCE

Applied Science Intern (Ads Measurement & Identity) | The Trade Desk — Jun 2026 – Sep 2026 | New York, NY

- Developed a **probabilistic audience-reach estimator** using censored negative-binomial models and identity-graph adjustments to account for limited identity coverage, audience overlap, and churn.
- Built **scalable SQL and PySpark data pipelines** to analyze **16B+ ad impressions**, leveraging existing HyperLogLog sketches to identify measurement discrepancies and generate ML-ready features for downstream modeling.
- Benchmarked reach estimates against **legacy models** and **independent ad-exposure signals**, using **A/B testing, temporal and entity holdouts,** and **sensitivity analyses** to evaluate model accuracy and reliability.
- Distilled offline statistical reach estimates into **LightGBM models**, achieving approximately **9% multiplier MAPE** against the offline reference with **sub-50 ms inference**.
- Partnered with product and engineering teams to define reach KPIs and integrate model outputs through **Databricks APIs and SQL dashboards**, translating cohort diagnostics into audience-planning decisions.

Data Scientist II (AI/ML Engineering) | Deloitte — Oct 2022 – Aug 2025 | Delhi, India

- Built a demand forecasting platform for **10K+ inventory parts** using ARIMA, Prophet, XGBoost, and LSTM, selecting the best-fit model per part and achieving **93.63% forecast accuracy (1 - MAPE)**, **52% fewer backorders**, and **$5M+ in savings**.
- Productionized end-to-end **ML training** and **batch-inference pipelines on AWS**, automating feature engineering, scheduled retraining, data-quality checks, and drift monitoring to reduce runtime by **91%**.
- Developed **Power BI KPI dashboards** tracking forecast accuracy, model performance, and operational exceptions, reducing reporting turnaround by **42%** and supporting inventory-planning decisions.
- Designed an explainable scoring and classification framework using feature-weighted aggregation, distribution-aware binning, Random Forest, and LightGBM, achieving **95% holdout classification accuracy** with drift-aware recalibration.
- Architected an **NLP standardization** and **defect-mining pipeline** for 2M+ records using BERT embeddings, HDBSCAN, and Apriori, mapping **31%** of descriptions to canonical labels and improving defect detection yield by **1.6×**.

PROJECTS & RESEARCH

Causal Reasoning & LLM Evaluation — Sep 2026 – Nov 2026

- Developed a **tool-augmented causal reasoning framework** using Python, NetworkX, and Pydantic to verify LLM conclusions through causal DAGs, intervention queries, backdoor adjustment, and evidence-sufficiency checks.
- Benchmarked **360 causal reasoning problems across 72 held-out graphs**, improving causal correctness from **68% to 84%**, reducing unsupported answers by **78%**, and limiting false abstentions to **8%** compared with direct LLM reasoning.

LLM-as-a-Judge for Summary Evaluation — Jan 2026 – Mar 2026

- Designed an **LLM-as-a-Judge evaluation framework** to assess AI-generated summaries against source documents using rubric-guided teacher LLM judgments across coherence, factual consistency, fluency, and relevance.
- Distilled teacher judgments into **Qwen2.5-3B using PyTorch** and **LoRA**, achieving **85.5% agreement with human ratings** across **1,200 held-out summaries** and outperforming a **zero-shot baseline by 12 percentage points**.

Personalized Ad Ranking for a Housing Marketplace — Oct 2025 – Dec 2025

- Built a **personalized listing** and **ad-ranking system** using **500K+ clickstream events** and **100K+ user profiles**, engineering behavioral features and user-item affinity scores to rank relevant content.
- Optimized **personalized recommendations** and **audience targeting** through behavioral segmentation, achieving **15% higher NDCG@10** than a popularity-based baseline and improving **planning-signal quality by 32%**.

TECHNICAL SKILLS

- **Programming & Libraries:** Python, SQL, R, Pandas, NumPy, scikit-learn, statsmodels, NLTK
- **Data Science & Experimentation:** Statistical Inference, Probabilistic Modeling, Causal Inference, A/B Testing, Experimental Design, Hypothesis Testing, Regression, Time-Series Forecasting, Cross-Validation, Calibration
- **Machine Learning, Deep Learning & LLMs:** XGBoost, LightGBM, Random Forest, PyTorch, TensorFlow, BERT, LoRA, RAG, LLM-as-a-Judge, Knowledge Distillation
- **Data Engineering & ML Systems:** PySpark, Databricks, AWS (S3, SageMaker, Step Functions), Model Deployment, Batch Inference, Feature Engineering, Model Monitoring, Drift Detection, Git
- **Visualization & Product Analytics:** Power BI, Tableau, Matplotlib, KPI Dashboards, Cohort Analysis, Product Metrics